

INTERNSHIP REPORT

Predicting Future Real Estate Prices in Tunisia Using Machine Learning

Host Organization: Société Tunisienne de Banque (STB)

Duration: Six-week internship

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Abstract

This report covers the six weeks I spent as part of a four-person intern team at Société Tunisienne de Banque, building a machine learning model to predict the future prices of houses and apartments in Tunisia three, four, and five years out.

We followed the CRISP-DM methodology from end to end: starting from a first raw dataset built by merging seven different sources into 42,161 rows across 28 columns, cleaning and imputing it down to a working set of 12,971 listings, engineering location and price-trend features, training and comparing several algorithms, and comparing different approaches using the same train/test split to ensure a fair evaluation.

The motivation is a practical one: STB, like any bank issuing mortgage loans, doesn't just need to know what a property is worth today it needs a defensible read on what it's likely to be worth over the life of the loan. A model that can project that forward, with an uncertainty interval to give an indication of prediction reliability.

The final solution consisted of two separate XGBoost models, one for villas and one for apartments, combined with a controlled price-trend index and quantile-regression confidence intervals reaches an R^2 of 0.806 on villas and 0.570 on apartments, with the 95% intervals covering the real price 93.9% of the time on data the models never saw during training.

Beyond the technical results, this internship was an opportunity to apply theoretical knowledge from my studies to a real business problem involving imperfect data, project constraints, and practical decision-making.

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List of Abbreviations

Abbreviation	Description
STB	Société Tunisienne de Banque
CRISP-DM	Cross-Industry Standard Process for Data Mining
ML	Machine Learning
AI	Artificial Intelligence
EDA	Exploratory Data Analysis
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
R ²	Coefficient of Determination
VIF	Variance Inflation Factor
SHAP	SHapley Additive exPlanations
XGBoost	Extreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine
TND	Tunisian Dinar
API	Application Programming Interface
JSON	JavaScript Object Notation
CSV	Comma-Separated Values
CV	Cross-Validation

HTML	HyperText Markup Language
CNSS	Caisse Nationale de Sécurité Sociale
ETAP	Entreprise Tunisienne d'Activités Pétrolières
CNAM	Caisse Nationale d'Assurance Maladie
BDET	Banque de Développement Economique de Tunisie
BNDT	Banque Nationale de Développement Touristique
SICAV	Société d'Investissement à Capital Variable
SICAR	Société d'Investissement à Capital Risque

General Introduction

The banking sector is increasingly adopting data science and artificial intelligence techniques to improve decision-making processes, and real estate valuation is one of the places where it matters most. For a bank like STB, which finances mortgages and real estate projects, knowing what a property is actually worth not just today, but over the years a loan will run directly shapes how it prices risk and structures financing.

That's the gap this internship tried to close. Over six weeks at Société Tunisienne de Banque, our four-person team built a machine learning model to predict future prices for houses and apartments across Tunisia. We used the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology to keep the project structured, to provide a structured framework throughout the limited internship period.

Concretely, the goal was a model that could estimate a property's current market value and then project that estimate 3, 4, or 5 years ahead, so STB could factor projected value not just today's asking price into mortgage, investment, and risk decisions.

This report follows the project in four chapters. Chapter 1 introduces STB. Chapter 2 lays out the context, the problem, and how our team organized the work. Chapter 3 walks through the full CRISP-DM methodology as we applied it. Chapter 4 covers the actual implementation tools, results, what went wrong, and possible improvements and future perspectives. It closes with a general conclusion, references, and appendices with the supporting detail.

Chapter 1: Presentation of the Host Organization

1.1 Introduction

Here, I introduce the company where I completed my internship, including its history, mission, organizational structure, and the department where I worked.

1.2 Presentation of Société Tunisienne de Banque (STB)

The Société Tunisienne de Banque (STB) is one of the major public commercial banks in Tunisia. It was founded in 1957 and has played an important role in financing the Tunisian economy ever since. The bank provides financial services to individuals, companies, and public institutions, and today operates one of the largest branch networks in the country.



Figure 1.1 — Société Tunisienne de Banque

Attribute	Description
Date established	January 18, 1957 (constituent assembly); began operating March 26, 1958
Founder	Abdelaziz Mathari
Legal form	Public limited company (Société anonyme), governed by law n°2016-48 of July 11, 2016 on banks and financial institutions
Share capital	776,875,000 Tunisian dinars
Shareholders	Tunisian State (71.54%) · CNSS (5.96%) · ETAP (2.22%) · CNAM (2.32%) · Others (18.18%)
Head office	Avenue Hédi Nouira, 1001 Tunis

Table 1.1 — Key facts about Société Tunisienne de Banque

1.3 History of STB

STB was born in the immediate aftermath of Tunisian independence. It began operating on March 26, 1958, as the first financial institution specifically built for the newly independent country. It's a response to a period when credit markets were disorganized and capital was flowing out of Tunisia rather than into it. It was set up as a mixed-economy company, with the Tunisian state holding a 52% stake from the start.

Over the following decades, STB was involved in a long list of firsts for the Tunisian banking sector: it opened the country's first public bank counter, launched the well-known "Livret Rouge" savings passbook in 1965, was the first Tunisian bank to introduce Visa card payments in 1978, and set up the country's first foreign-exchange trading room in 1992. It also founded Sicav l'Epargnant in 1991 and the country's first SICAR (venture capital investment company) in 2001.

The most significant turning point for the bank's current structure came in 2000, when STB merged with two state-owned development banks the Banque de Développement Economique de Tunisie (BDET) and the Banque Nationale de Développement Touristique (BNDT) becoming one of the largest banking groups in the Maghreb and on the African continent. Between 2016 and 2020, STB carried out a further transformation programme, modernizing its core information system and its digital payment infrastructure, which is part of the same broader push toward data-driven decision-making that this internship project fits into.

1.4 STB's Mission and Vision

STB describes itself as a universal bank meaning it combines retail banking, corporate banking, investment banking, and development financing under one roof, rather than specializing in just one of these. Since its creation, the bank's role has gone beyond a purely commercial one: it has consistently positioned itself as a driver of Tunisia's economic and social development, historically taking the lead on major national projects in industry, tourism, agriculture, and trade.

Today, that same orientation continues through STB's ongoing digital transformation strategy, which aims to modernize customer-facing services (mobile and online banking, bancassurance, digital payments) while also using data and technology internally to support faster, more consistent decision-making including in areas like credit risk assessment and, more specifically to this report, property valuation for mortgage lending.

1.5 Main Activities and Services

STB provides a broad range of banking and financial services for individuals, companies, and institutions. Its main activities include:

- **Retail banking:** account management, savings solutions, payment services, and personal loans for individuals.
- **Corporate banking:** business loans, project financing, and support for companies managing their day-to-day financial activities.
- **Mortgage financing:** loans for purchasing, building, or renovating real estate. This is the activity most directly connected to this internship

project, since it depends on accurate property valuation to assess risk and set financing conditions.

- **Investment banking:** support for investment projects and financial solutions for businesses and organizations.
- **International trade finance:** support for import/export companies, including documentary credit and international transaction facilitation.
- **Digital banking services:** remote access to banking operations through digital platforms, improving accessibility and customer experience.
- **Risk management:** identifying, evaluating, and controlling credit and financial risk to support safe, effective decision-making across the bank.

1.6 Organizational Structure

STB is organized into several departments that manage its various activities, including Finance, Information Systems, Risk Management, Human Resources, Marketing, and Data Analysis. Each department contributes to the bank's day-to-day operations and to its broader strategic objectives.

1.7 Presentation of the Internship Department

I completed my internship in the Data Department. This department is responsible for developing digital solutions, managing databases, supporting business intelligence projects, and building data-driven tools to improve the bank's efficiency which is exactly where a project like this one, predicting future property values from real estate data, naturally belongs.

1.8 Conclusion

This chapter gave an overview of STB: where it came from, what it does today, and the department this internship took place in. With that context in place, the next chapter turns to the internship itself the problem we were asked to look into and how our team approached it.

Chapter 2: Project Context and Problem Definition

2.1 Introduction

The following sections explain the context of the project, the problem we were asked to solve, our objectives, how our team was organized, and how the work was planned over the six-week internship.

2.2 Internship Context

During this internship, STB showed interest in exploring data-driven approaches for improving real estate valuation processes. One concrete challenge the bank faces is estimating the future value of real estate properties used as collateral for mortgage loans.

Traditional valuation methods mostly rely on expert judgment and current market conditions at the time of the loan. Predicting how a property's value might change over the next few years, however, requires something different: analysing historical trends and using machine learning to pick up on patterns that aren't obvious from a single snapshot in time.

2.3 Problem Statement

Banks need reliable tools to help predict future real estate prices, both to improve risk assessment and to support investment decisions.

The main question we were asked to address during my internship was: *how can machine learning techniques be used to predict the future prices of houses and apartments over the next 3, 4, and 5 years?*

2.4 Project Objectives

The main objective was to develop a machine learning model capable of predicting future residential property prices. More specifically, we set out to:

- Collect real estate data from different sources.

- Preprocess and clean the collected data.
- Explore and analyse the dataset.
- Create relevant features to improve prediction.
- Train different machine learning models.
- Compare the performance of the models.
- Select the best-performing model.
- Evaluate the accuracy of the predictions.
- Provide a tool that can support STB in real estate valuation.

2.5 Project Scope

To keep the project realistic for a six-week internship, we deliberately kept the scope narrow in a few ways:

- Geographic scope: Tunisia only, using listings spread across the country's governorates rather than a single city or region.
- Property types: residential apartments and villas only. Land, offices, commercial premises, and short-term vacation rentals were excluded from the start, both from the data we collected and from the models we trained.
- Forecast horizon: price projections at 3, 4, and 5 years ahead, built on top of a current value estimate rather than as an independent prediction.
- Data sources: publicly available listing data (scraped or pre-collected), not STB's internal loan or property records, which we did not have access to. This means the model reflects asking prices on the open market rather than actual bank-appraised values, which is an important distinction discussed further in Chapter 4.

2.6 Team Organization

The project was carried out by a team of four interns. We worked through the CRISP-DM phases: data collection, data preprocessing, model development, evaluation, and documentation.

Thanks to regular communication between team members, plus check-ins with our internship supervisor, kept the project on track and gave us a chance to

catch problems early rather than after they'd already shaped the rest of the pipeline.

2.7 Project Planning

To fit everything into six weeks, we built a plan around the CRISP-DM phases. We met with our supervisor regularly once a week to review progress, flag anything that wasn't working, and validate results before moving to the next phase.

Week	Activities
Week 1	Understanding the project objectives, studying the CRISP-DM methodology, and organizing the work between team members.
Week 2	Collecting real estate data and performing an initial analysis of the dataset.
Week 3	Cleaning and preprocessing the data to prepare it for the modeling phase.
Week 4	Selecting machine learning algorithms and training different prediction models.
Week 5	Evaluating the models, comparing their performances, and selecting the best model.
Week 6	Finalizing the project, writing the report.

Table 2.1 — Six-week project planning

2.8 Conclusion

This chapter explained the context of the project, the problem we were asked to solve, our objectives, and how the work was organized throughout the internship. With this context established, the next chapter introduces the CRISP-DM methodology and explains how it guided the different stages of our project.

Chapter 3: Project Methodology (CRISP-DM)

3.1 Introduction

CRISP-DM (Cross-Industry Standard Process for Data Mining) is one of the most widely used methodologies for structuring a data mining or machine learning project. It breaks the work down into six phases. On paper they look sequential, but in practice we went back and forth quite a bit a problem discovered during modeling would often send us back to fix something in data preparation, and sometimes even back to data understanding. This chapter describes what each phase meant for our specific project; the concrete implementation details and the actual results are presented separately in Chapter 4.

- **Business Understanding:** figure out what the project needs to achieve from the bank's point of view.
- **Data Understanding:** get the raw data together and look closely at what's actually in it.
- **Data Preparation:** clean it up and turn it into something a model can be trained on.
- **Modeling:** pick one or more algorithms, train them, and tune their parameters.
- **Evaluation:** check whether the models meet the objectives set out at the start.
- **Deployment:** package everything so it can be used, and document what it can and can't do.

3.2 Business Understanding

There isn't currently a standardized, data-driven way to estimate property values in Tunisia. Buyers, sellers, agents, and lenders mostly rely on manual comparisons with similar properties, which works reasonably well but is inconsistent and doesn't scale. This project explores whether a model trained on

scraped listing data can produce reliable valuations, and whether those valuations can reasonably be projected a few years into the future.

3.2.1 Business Objectives

- **Primary objective:** estimate the current market value of a residential property (apartment or villa) in Tunisia based on its features and location.
- **Secondary objective:** push that estimate forward in time, N years ahead, for situations where the future value matters more than today's price for example, when a property is used as collateral for a long-term loan.

3.2.2 Success Criteria

- **Business criteria:** the estimates should be good enough to serve as a useful starting point for a human valuation not a replacement for one and should come with a confidence range that honestly reflects how the model performs (overall R^2 around 0.57; villas between 0.73 and 0.81 depending on the model version; apartments between 0.54 and 0.57).
- **Technical criteria:** a test-set R^2 clearly above a naive baseline, reasonable performance on both apartments and villas, and ideally a forecasting component checked against real historical outcomes rather than just assumed to work.

3.2.3 Project Requirements

- A dataset large and varied enough to represent the Tunisian residential market, covering both apartments and villas across multiple governorates.
- A price trend signal, since forecasting future value requires knowing how prices have moved historically, not just what a property is worth today.
- An uncertainty estimates alongside every prediction, given that this model would be used to support financial decisions rather than to replace human judgment outright.
- An explicit acknowledgment that villa data would likely be too limited to build a fully independent villa-specific price trend, meaning some assumption of shared behaviour with apartments would probably be needed.

3.3 Data Understanding

3.3.1 Data Collection

Tunisia doesn't have a public, structured database of real estate listings, so the first real task of the project was simply finding data and putting it together from multiple places. We ended up combining seven different sources: five ready-made files we found and downloaded, and two we scraped ourselves directly from Tayara.tn and Mubawab.tn, two of the most popular Tunisian classifieds sites for real estate.

For Tayara.tn, rather than parsing rendered HTML pages, our scraper calls the same internal JSON API the site itself uses (its Next.js data endpoints), which returns cleaner structured data. Listings are filtered immediately to keep only apartments and villas, using the subcategory IDs Tayara assigns internally to each property type. For Mubawab.tn, we scraped more conventionally, requesting the site's "apartments for sale" and "houses for sale" category pages governorate by governorate and parsing the HTML with BeautifulSoup. Both scrapers use a normal browser user-agent and a polite delay between requests.

Within the internship timeframe, these two scraping runs brought in a modest number of listings 77 from Tayara and 164 from Mubawab much smaller than the older, pre-existing files, but still useful since they're the most recent listings in the dataset and the only ones we collected and controlled end to end.

3.3.2 Data Sources

The full list of sources merged into the raw dataset is detailed in Chapter 4 (Table 4.1), together with the fill rate of each. In summary: a 2026 property-price export, a 2025 property-price dataset, a 2024 apartment-sales file, a general real-estate dataset, an English-language Tunisia real-estate file, and our two scraped sources (Tayara and Mubawab). Each source had its own structure, column names, and language (French or English), which meant every one of them needed its own mapping before it could be combined with the others.

3.3.3 Data Description

We defined one common target schema 26 columns covering property type, price, location, surface area, room counts, and amenities and wrote a mapping dictionary for each source translating its own column names into that shared

schema. Some columns didn't correspond to anything useful and were dropped outright: a pre-computed log-price column (redundant with price itself), raw address strings, listing URLs, and scraped phone numbers. Once every source had been reshaped into the same format and stacked together, the result was a first combined dataset of 42,161 rows across 28 columns.

The file used for modeling later in the project, "df_all_properties_apres_apiregex.xls", is smaller 15,408 rows. The name itself ("after Api/regex") where we extract structured values from free-text titles and descriptions.

3.3.4 Exploratory Data Analysis (EDA)

Our initial look at the modeling dataset focused on two things: how complete each column was, and how listings were distributed geographically. Fill rates on the core structural columns were low and uneven living area was 99.8% complete, but room and bathroom counts ranged from 37% to 49%, and several amenity flags were filled in less than 1% of the time. Geographically, the data is heavily concentrated around Greater Tunis, which made sense given where most listing activity happens, but it also meant some governorates had very little data to work with. Both findings directly shaped the data preparation choices described next. The detailed figures and tables for this exploration are presented in Chapter 4, Sections 4.4 and 4.5.

3.4 Data Preparation

This was, by a wide margin, the most time-consuming phase of the project.

3.4.1 Data Cleaning

We removed listings with clearly wrong values living areas under 15 m², prices per m² outside a sane range, and rows with no price at all and, once the dataset was fully cleaned and imputed, removed exact duplicate rows to prevent the same listing from appearing in both the training and test sets.

3.4.2 Handling Missing Values

Room counts (rooms, bedrooms, bathrooms) were rebuilt through a chain of cross-imputation steps between related columns, with a fallback based on living-area bins where nothing else was available. Floor count was imputed

similarly. Amenity flags (pool, garden, garage, air conditioning, security, terrace) couldn't be learned reliably from the data, so missing values were set to 0 (absence assumed), with one logical rule layered on top: a property with a pool was assumed to also have a garden. Governorate and city names were normalized and cross-repaired against each other, and land area for villas was imputed using governorate-level medians.

3.4.3 Feature Engineering

On top of cleaning, we engineered a number of new features: a postal code derived from an exhaustive lookup of Tunisian localities, a broader region grouping, a tourism-zone flag based on well-known tourist towns, and most importantly for the forecasting component a controlled price-trend index built from apartment listings, used to bring prices from different years onto a common scale (deflated to a 2018 base year) before training.

3.4.4 Data Transformation

Governorate, with a manageable 25 categories, was one-hot encoded. City, with roughly 265 categories, was encoded using K-Fold target encoding instead, fitted only on the training folds to avoid leaking test-set price information into the encoding. Prices were deflated using the trend index described above before being used as the model's training target.

3.4.5 Dataset Splitting

The data was split 80/20 into training and test sets, stratified by property type so both sets kept a realistic mix of villas and apartments. This split was then reused consistently across every model comparison in the project, so results could be compared fairly against each other.

3.5 Modeling

3.5.1 Selection of Machine Learning Algorithms

We started with a linear baseline (Ridge Regression) to get a sense of how linear the pricing problem is, then moved to gradient-boosted tree models XGBoost, and later HistGradientBoosting and LightGBM for comparison. Since tree-based methods tend to handle the kind of non-linear interactions and mixed

categorical/numeric features common in real estate data. A Dummy regressor (always predicting the mean) was included purely as a sanity-check baseline.

3.5.2 Model Training

Beyond the baseline comparison, we trained models in two different configurations: a single combined model covering both villas and apartments, and after comparing results two separate models, one per property type, each with its own tailored feature set. A separate quantile-regression model was also trained to produce prediction intervals rather than single point estimates.

3.5.3 Hyperparameter Tuning

We used a randomized search (RandomizedSearchCV, 50 iterations, 5-fold cross-validation) over XGBoost's main parameters number of trees, tree depth, learning rate, and the two sampling rates to avoid the cost of an exhaustive grid search while still covering a reasonably wide parameter space.

3.6 Evaluation

3.6.1 Evaluation Metrics

Three complementary metrics were used throughout: RMSE (Root Mean Squared Error), which penalizes large errors more heavily; MAE (Mean Absolute Error), which is easier to interpret directly in TND; and R^2 (coefficient of determination), which expresses how much of the price variation the model explains, relative to always predicting the average price.

3.6.2 Model Comparison

Every model was compared on the exact same train/test split, so differences in the numbers reflect real differences in the models rather than differences in how the data happened to be divided. Beyond the initial algorithm comparison, we also ran comparisons for specific ideas added location features, and villa/apartment segmentation always against the same reference split.

3.6.3 Selection of the Best Model

The final architecture was chosen based on segment-level performance rather than a single overall number, since a configuration that looked fine on average could still be quietly underperforming on one segment. This is what led to the

decision to train two separate models with different feature sets rather than one model for everything the reasoning and the numbers behind that decision are presented in full in Chapter 4.

3.7 Deployment (Proposed)

The intended deployment is a single function, `forecast_price(property_features, years_ahead)`, that routes a property to the correct model (villa or apartment), derives its location features automatically, returns a point estimate together with a 95% confidence interval, and applies the trend index to project the estimate forward in time. The models, encoders, and reference tables needed to run this function outside the original notebook were exported so a simple test interface, shown in Section 4.4.9, was built on top of it. This remains a proposed deployment rather than a production one STB has not yet integrated it into any live system.

3.8 Conclusion

With the methodology established, the next chapter focuses on how it was implemented in practice, the results we obtained, and the lessons learned throughout the project.

Chapter 4: Implementation and Results

4.1 Development Environment

This chapter presents the practical implementation of the project, the tools used, the experimental results, and the challenges encountered during development

4.2 Tools and Technologies

- **Data handling:** pandas and NumPy for all data manipulation, cleaning, and feature engineering.
- **Web scraping:** the requests library together with BeautifulSoup for Mubawab, and direct calls to Tayara's internal JSON API.
- **Modeling:** scikit-learn (Ridge Regression, preprocessing, cross-validation, RandomizedSearchCV), XGBoost, LightGBM, and scikit-learn's HistGradientBoostingRegressor.
- **Model interpretation:** SHAP (TreeExplainer) for feature importance, and statsmodels for the VIF collinearity check.
- **Persistence and deployment:** joblib for exporting trained models and encoders, and Streamlit for a lightweight test interface.
- **Visualization:** matplotlib and seaborn for the exploratory and evaluation charts referenced throughout this chapter.

4.3 Implementation Process

In practice, the pipeline ran in the following order: collect and merge the seven raw sources into a single file; clean and impute the merged dataset; deduplicate and build the price trend index; split the data into training and test sets; train and compare baseline models; tune the best-performing algorithm; test additional features and the villa/apartment segmentation; settle on a final two-model architecture; and finally wrap everything into the `forecast_price()` function and export the artifacts needed to run it independently. The sections below walk through the concrete results produced at each of these stages.

4.4 Experimental Results

4.4.1 Building the Raw Dataset

As described conceptually in Chapter 3, the raw dataset was assembled from seven sources. Table 4.1 shows each source together with its average fill rate once mapped into the shared schema generally between 30% and 37%, which is low but expected, since no single source fills in every field and stacking sources with different column sets naturally creates gaps.

Source	What it is	Average Fill Rate
df_2026	2026 property-price export	33.3%
df_property_prices2025	2025 property-price dataset (apartments and villas only)	33.3%
df_apartmentsales2024	2024 apartment sales file	37.0%
df_mubawab	Scraped from Mubawab.tn	33.6%
df_tayara	Scraped from Tayara.tn (internal API)	36.4%
df_dataset	General real-estate dataset	29.6%
df_tunisia_real_estate	Tunisia real-estate file (English columns)	33.3%

Table 4.1 — The seven sources merged into the combined raw dataset (42,161 rows, 28 columns)

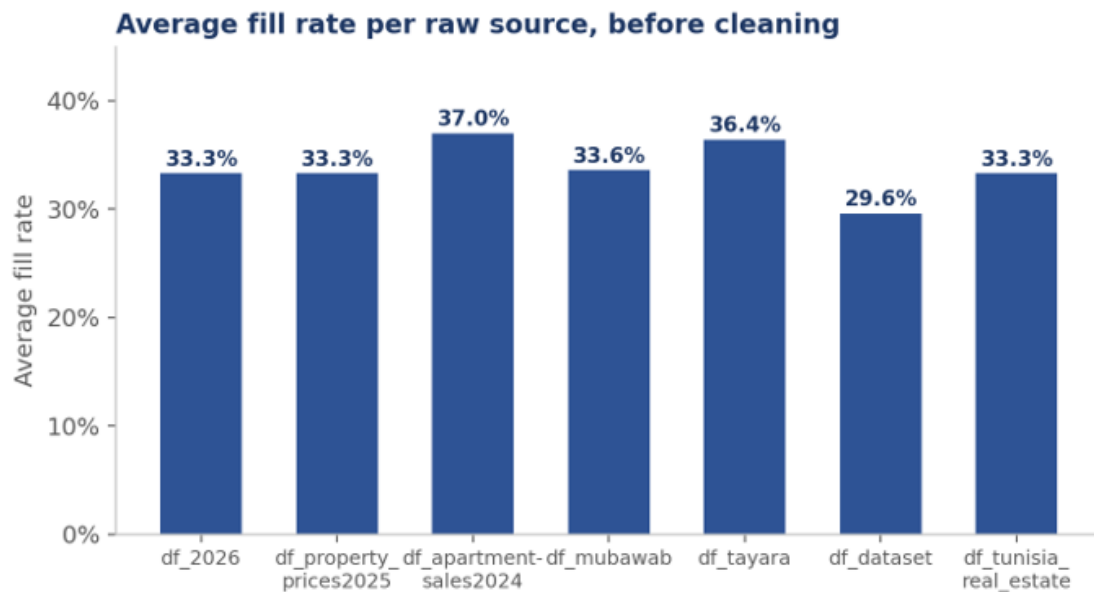


Figure 4.1 — Average fill rate per raw source, before cleaning.

4.4.2 Fill Rates in the Modeling Dataset

Table 4.2 and Figure 4.2 show the fill rate of the core structural columns before any cleaning took place.

Column	Initial Fill Rate	Comment
surface_habitable_m2 (living area)	99.8%	Essentially complete
nb_pieces / nb_chambres / nb_salles_bain (rooms/bedrooms/bathrooms)	37% / 49% / 49%	Needed a fair amount of cascading imputation
nb_etages (number of floors)	Low	Imputed from surface- area bins
Amenities: pool, garden, garage, air conditioning, security	< 1%	Almost entirely missing at the source

terrasse (terrace)	34 values out of 15,408	Every single non-missing value was a "1"
etat_bien, annee_construction (condition, year built)	0%	Unusable, dropped entirely
id_annonce (listing ID)	1.3%	Not reliable as a unique identifier

Table 4.2 — Fill rates of the main columns before cleaning, on the 15,408-row modeling dataset

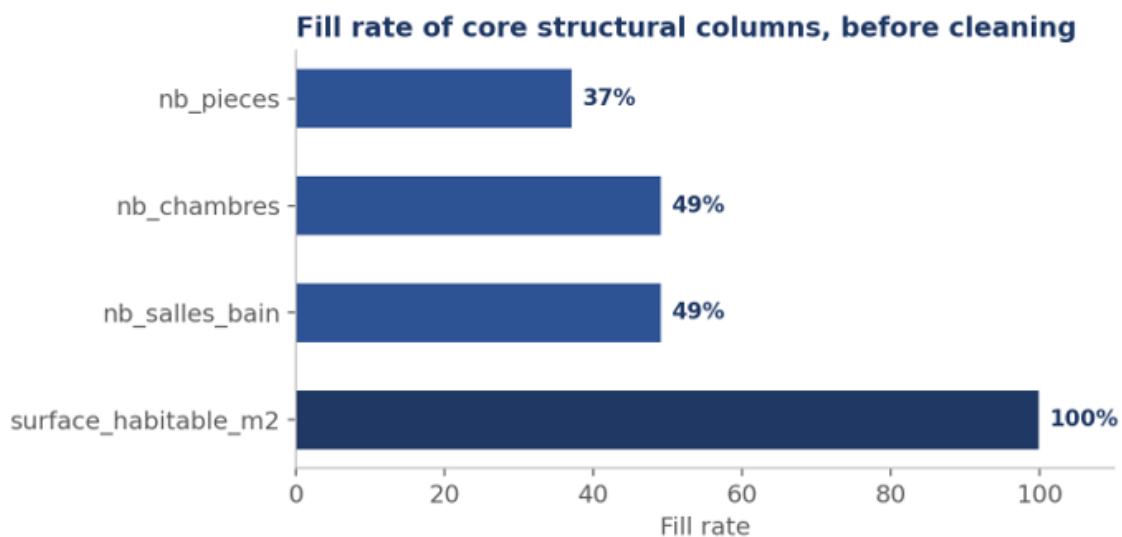


Figure 4.2 — Fill rate of core structural columns, before cleaning.

4.4.3 Data Cleaning and Imputation — What Actually Happened

Room counts were rebuilt using the modal offset between rooms and bedrooms per source, with a living-area-bin fallback, repeated iteratively so newly filled values could help fill in whatever was still missing elsewhere. A consistency check found 257 rows where bedroom count exceeded room count, which were corrected by capping. Floors were imputed from surface-area bins per property

type. Amenities, having shown no learnable relationship with price or size, were filled with 0. Governorate and city names were normalized and cross-repaired, and villa land area was imputed by governorate median (597 m² as the overall fallback).

Filtering Step	Rows Removed	Rows Remaining
Starting point	42,161	15,408
Living area under 15 m ²	50	15,358
Price per m ² outside a sane 50–15,000 TND/m ² range	172	15,186
Missing price	26	15,160

Table 4.3 — Filtering out outliers and incomplete records, step by step

During the validation process, an inconsistency was identified in an intermediate row-count summary. This issue was related to manual tracking and did not affect the final dataset, whose values were recalculated directly from the dataframe.

After all of this, we were left with a clean dataset of 12,971 rows.

4.4.4 Deduplication and the Price Trend

Before training, we ran a strict deduplication, dropping rows identical across every column. That removed 2,189 listings out of 15,160 (about 14.4%), almost certainly duplicate scrapes of the same listing.

Step	Rows Before	Rows After
------	-------------	------------

Full-row deduplication	15,160	12,971
------------------------	--------	--------

Table 4.4 — Effect of the final deduplication pass on the modeling dataset

We then built a controlled price trend by regressing $\log(\text{price per m}^2)$ on publication year, governorate, living area, and room count, fitted only on apartment listings, and used it to bring every listing's price back to a 2018 base year.

Year	Controlled Price per m ² (TND)
2018 (base year)	1,238
2019	1,259
2020	1,281
2021	1,304
2022	1,326
2023	1,350
2024	1,373
2025	1,397
2026	1,421

Table 4.5 — Controlled price-per-m² trend index (apartment listings only) — roughly +1.6% per year

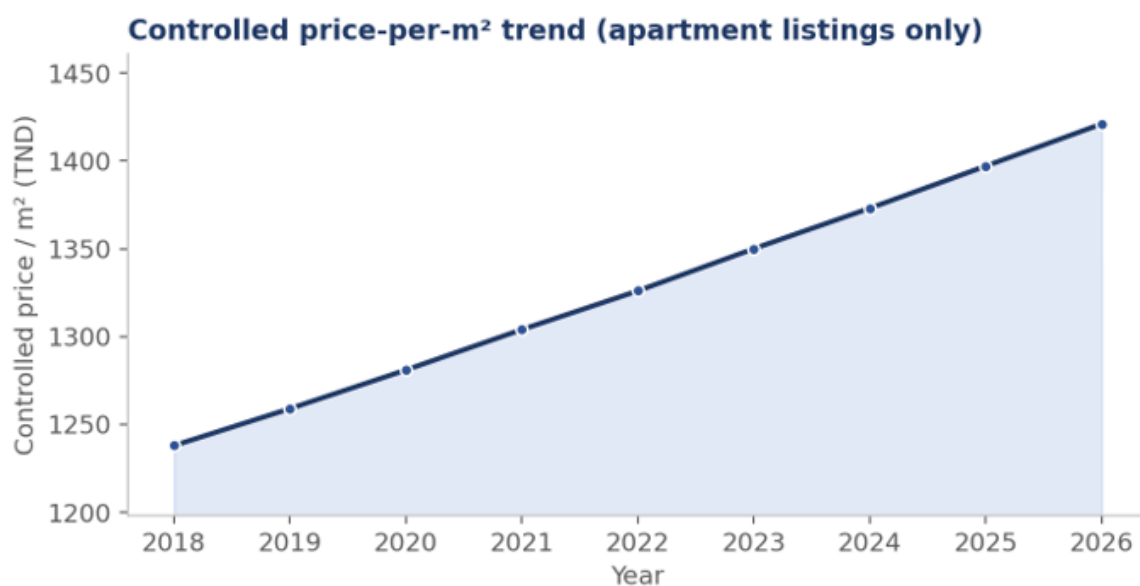


Figure 4.3 — Controlled price-per-m² trend (apartment listings only).

4.4.5 Baseline Model Comparison

We split the data 80/20, stratified by property type (10,376 training rows, 2,595 test rows), one-hot encoded governorate, and K-Fold target-encoded city. A first comparison between Ridge and untuned XGBoost showed a large gap:

Model	RMSE (TND)	MAE (TND)	R ²
Ridge Regression	631,674	250,042	0.121
XGBoost (untuned)	438,838	161,334	0.576

Table 4.6 — Ridge vs. XGBoost, single combined model (villa + apartment)

Broken down by segment, untuned XGBoost reaches $R^2 = 0.738$ on villas and $R^2 = 0.546$ on apartments. Ridge simply isn't flexible enough to capture the kind of interactions that seem to matter for real estate prices.

4.4.6 Tuning and Extended Comparison

A randomized hyperparameter search (50 iterations, 5-fold cross-validation) found the following best XGBoost configuration:

- `max_depth = 5`
- `n_estimators = 252`
- `learning_rate  $\approx$  0.039`
- `subsample  $\approx$  0.60`
- `colsample_bytree  $\approx$  0.60`

Interestingly, the tuned model achieved a slightly lower R^2 than the default XGBoost model (0.567 vs. 0.576). We kept the tuned version as our reference model regardless, since the gap is small and the search itself was cross-validated rather than the result of a single lucky split.

We extended the comparison with two more gradient-boosting algorithms to make sure XGBoost wasn't an arbitrary choice, and included a Dummy baseline as a sanity check:

Model	RMSE (TND)	MAE (TND)	R^2
Dummy (mean prediction)	673,968	331,143	-0.000
Ridge Regression	631,674	250,042	0.121
XGBoost (tuned)	443,633	164,913	0.567
HistGradientBoosting	443,387	163,092	0.567
LightGBM	439,763	163,558	0.574
XGBoost (untuned)	438,838	161,334	0.576

Table 4.7 — Full comparison, including a Dummy baseline and two additional algorithms

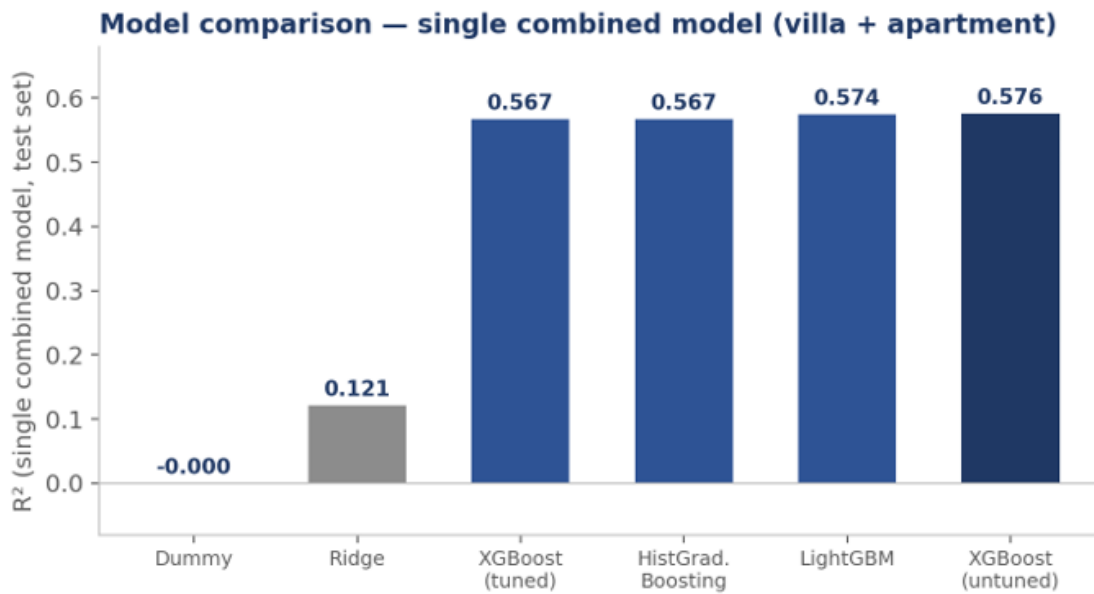


Figure 4.4 — Model comparison, single combined model (villa + apartment).

Leaving Ridge aside, the four gradient-boosting variants all land in a narrow band, R^2 between 0.567 and 0.576 suggesting the current ceiling is set more by what the data can tell us than by the choice of algorithm. A 5-fold cross-validation on the full dataset gave a mean R^2 of 0.593 (± 0.075), confirming the result is stable rather than a lucky split.

4.4.7 Uncertainty Intervals via Quantile Regression

A fixed-margin approach ($\pm 1.96 \times$ residual standard deviation) proved a poor fit for real estate prices, since errors are skewed and roughly scale with price. We trained a second XGBoost model with a quantile loss to predict the 2.5th, 50th, and 97.5th percentiles directly.

Calibration Metric	Measured Value
How often the 95% interval actually contained the real price (test set)	93.9% (target: 95%)
Median interval width	534,191 TND
Rows where the quantiles came out in the wrong order	8 of 2,595 (0.3%)

Table 4.8 — Calibration of the quantile-regression confidence intervals

4.4.8 Feature and Architecture Improvements

We tested a more exhaustive postal-code lookup (4,868 localities, 98.0% match rate), plus region and tourism-zone flags, added to the single combined model:

Region	Number of Listings
Greater Tunis	8,218
Cap Bon	1,918
Sahel	1,623
North	476
Sfax	335
South	243
Center	91
Other	67

Table 4.9 — Listing distribution by region (out of 12,971 rows after deduplication)

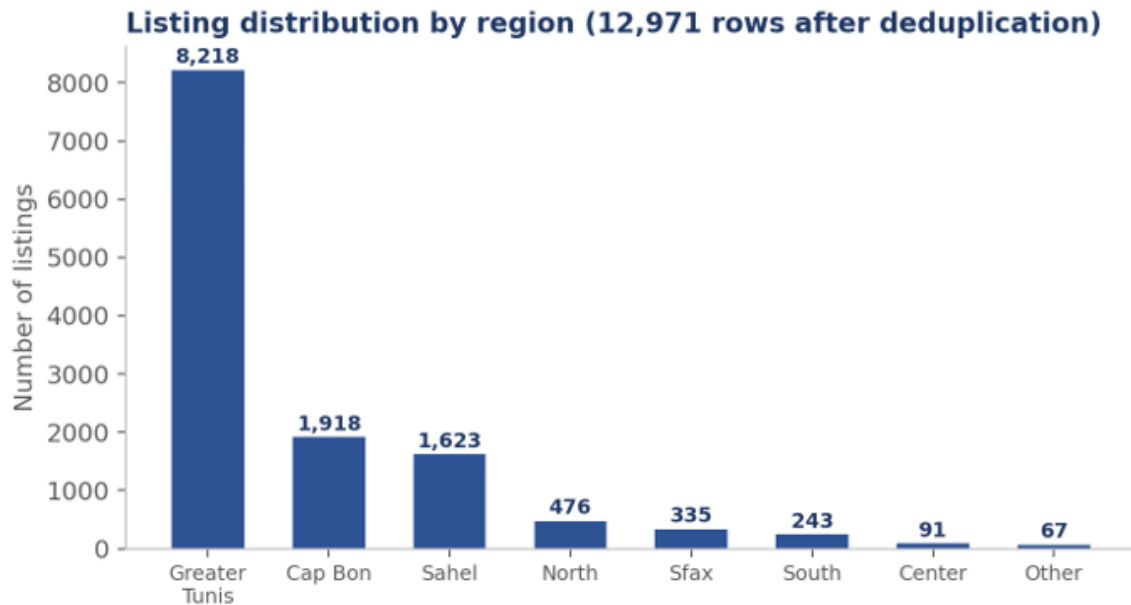


Figure 4.5 — Listing distribution by region.

Model	RMSE	MAE	Overall R ²	Villa R ²	Apartment R ²
XGBoost (governorate + city only — reference)	443,633	164,913	0.567	0.734	0.536
XGBoost + postal code + region + zone	444,534	164,512	0.565	0.641	0.550

Table 4.10 — Effect of adding richer location features to the single combined model

This one surprised us. Adding these extra location details clearly hurt the villa segment (R^2 dropping from 0.734 to 0.641), most likely because there aren't many villa listings and the 4,868-locality lookup creates a lot of rare categories the model can overfit to. Apartments improved slightly (0.536 to 0.550). That contrast is what pushed us toward testing full segmentation.

Segment	Single Model — MAE / R ²	Separate Models — MAE / R ²
Villa	147,219 TND / 0.734	135,196 TND / 0.806
Apartment	170,863 TND / 0.536	171,894 TND / 0.558

Table 4.11 — Single model vs. separate segment-specific models

Segmentation was a clear win for villas (R^2 0.734 $\rightarrow$ 0.806) without meaningfully hurting apartments. Combining both findings, we settled on two separate models with tailored feature sets:

Final Model	R ² (Test Set)	Test Set Size
Villa (without postal code / region)	0.806	653 listings
Apartment (+ postal code, region)	0.570	1,942 listings

Table 4.12 — Performance of the final two-model architecture

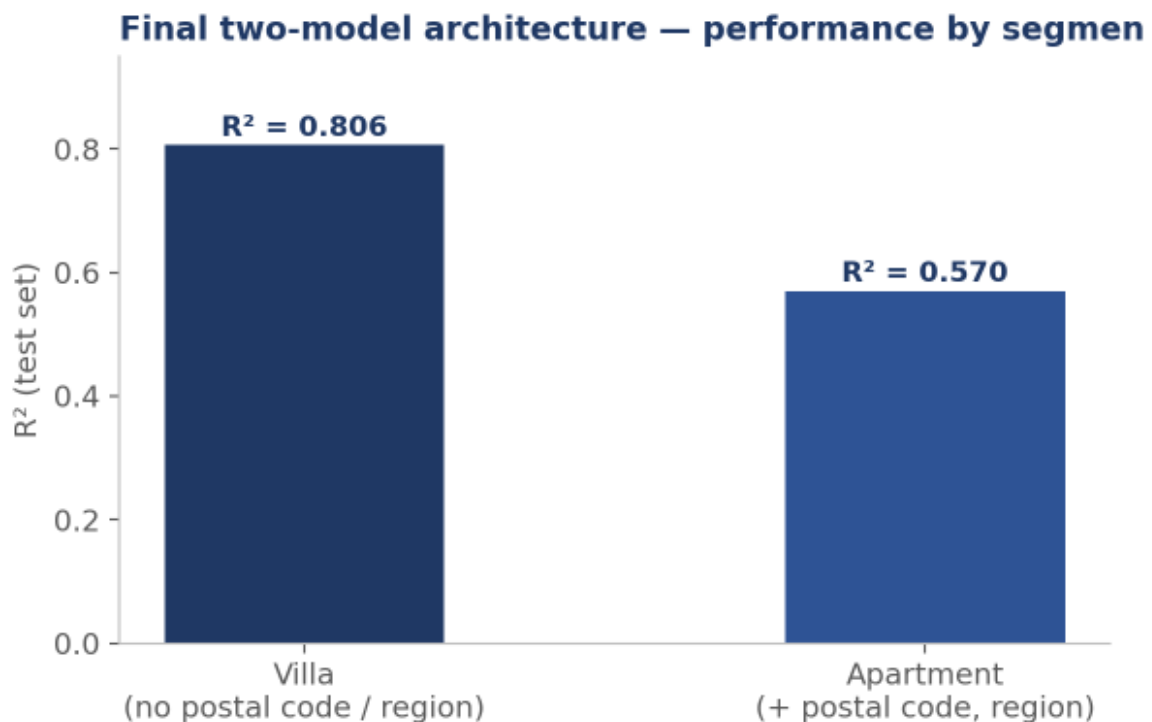


Figure 4.6 — Final two-model architecture — R² by segment.

4.4.9 Deployment Demonstration

The full pipeline was wrapped into `forecast_price(property_features, years_ahead)`. As a demonstration, a 120 m² apartment with 3 rooms, 2 bedrooms, and one bathroom, in Menzah 9 (Ariana), projected 5 years ahead to 2031, gives:

Point Estimate	Lower Bound (2.5%)	Upper Bound (97.5%)
305,313 TND	136,756 TND	522,704 TND

Table 4.13 — Sample output of `forecast_price()` for a typical apartment

Unlike the single-model version discussed in Section 4.4.7, this projection now draws its interval from two separate quantile models one trained on villas, one on apartments each using the same features and hyperparameters as its segment's point-estimate model. The quantile-crossing safeguard from 4.4.7 is still in place as a backstop, but for this example it doesn't need to do anything the interval comes out clean on both ends.

`forecast_price()` on its own is just a Python function, so we wrapped it in a small Streamlit app so it could actually be tested instead of only called from inside the notebook. The two models, their two quantile models, the four encoders, the trend index, and a small reference table of governorates, cities, and typical value ranges are exported once from the notebook and loaded when the app starts. The interface itself is split into two panels: a form on the left where a property's attributes are entered type, location, surface, room counts, amenities and a results panel on the right showing the point estimate, the 95% interval, and a small chart of how the projected price moves year by year out to whichever horizon was chosen. Figure 4.7 shows the interface running the same example as Table 4.13.

Estimation de prix immobilier

TUNISIE - MODELE XGBOOST VILLA / APPARTEMENT

Type de bien

appartement

Terrain associe ?

☐

Gouvernorat

Ariana

Surface terrain (m2)

0

Piscine

☐

Ville

Menzah 9

Garage

☐

Surface habitable (m2)

120

Jardin

☐

Nombre de pieces

3

Climatisation

☐

Nombre de chambres

2

Terrasse

☐

Salles de bain

1

Securite

☐

Nombre d'etages

1

Année de reference

2026

Projection (annees)

5

Estimer le prix

ESTIMATION POUR 2031

305 313 TND

Intervalle 95% : 136 756 - 522 704 TND

Figure 4.7 — Deployment interface (Streamlit), showing the example from Table 4.13.

4.5 Performance Analysis

Residual analysis by governorate flagged Jendouba as an outlier area, with a mean residual around 1.58 million TND likely a small-sample effect rather than anything specific to that governorate. SHAP analysis on the tuned model confirmed that surface area and location dominate the price prediction, with land area mattering far more for villas than apartments, as expected. A VIF check on the Ridge feature set showed the collinearity you'd expect from one-hot encoded governorate dummies (Tunis ≈ 37.9 , Ariana ≈ 34.9), which is normal and doesn't affect the tree-based models used for prediction.

4.6 Challenges Encountered

- Merging seven differently shaped data sources into one schema took far longer than we expected, mostly because every source used different column names, languages, and sometimes different units or conventions for the same underlying field.
- The two scrapers (Tayara, Mubawab) only brought in a small number of listings within the internship's timeframe, both because of the polite rate-limiting we built in on purpose and because we simply didn't have weeks to spare purely on scraping.
- Room-count imputation needed several iterative passes before the fill rate was acceptable a single pass of cross-imputation wasn't nearly enough given how sparse the original data was.
- The villa segment consistently behaved differently from the apartment segment throughout the project, which meant several assumptions that worked fine for apartments (like adding fine-grained postal codes) actively hurt performance on villas.
- The quantile-regression model and the point-estimate model, being trained independently, occasionally disagreed enough to produce an invalid interval a problem we hadn't anticipated until we saw it in the test output, which is why the clamping safeguard exists.
- We found and had to be upfront about a couple of inconsistencies in our own intermediate logging (a negative row count in one printed summary, and a duplicate-rate figure in an earlier note that didn't match a later re-run of the pipeline) a reminder that documentation needs to be re-checked every time the underlying code changes, not just written once.

4.7 Limitations of the Project

- **Villa intervals used to run a bit narrow:** in an earlier version of the model (before splitting villas and apartments), the villa confidence interval covered the real price about 91% of the time against a 95% target likely a side effect of how little villa data is available.
- **The villa trend isn't really a villa-specific trend:** there isn't enough villa history to fit its own trend, so the project borrows the apartment trend and assumes it applies a reasonable but unverified assumption.

- **Two potentially useful columns were unusable:** property condition and year built would normally be strong predictors but had a 0% fill rate in what we collected.
- **The model reflects asking prices, not appraised values:** since the data comes from public listings rather than STB's own records, the model estimates what sellers are asking, which is not necessarily identical to a bank-appraised market value.
- **The tourism-zone flag and trend/price scaling are dataset-specific:** the tourism-zone list is a manual approximation, not an official zoning boundary, and the trend/deflation is fitted to this specific dataset's sources and years rather than being universally applicable.

4.8 Future Improvements

- Backtest the forecasting component against real historical price outcomes, rather than relying purely on trend extrapolation.
- Extend the scraping effort over a longer period to collect more recent, higher-quality listings particularly for villas, where the sample size is currently the main bottleneck.
- Explore whether STB's own historical mortgage or appraisal records (subject to appropriate data governance) could complement or validate the public listing data used here.
- Revisit the villa price trend once enough multi-year villa data becomes available, rather than continuing to borrow the apartment trend.
- Move the proposed deployment from a Streamlit test interface toward a properly integrated internal tool, with monitoring and periodic retraining as new listing data comes in.

4.9 Conclusion

This chapter presented the implementation of the project, the experimental results, and the challenges encountered throughout the development process. Together with the methodology described in the previous chapter, it provides a clear understanding of the model's performance, its limitations, and the lessons learned during the internship.

General Conclusion

This internship provided an opportunity to develop a complete machine learning project in a real professional environment at Société Tunisienne de Banque. Starting from multiple heterogeneous real estate datasets, our team worked through the different stages of a data science pipeline, including data collection, preprocessing, feature engineering, model development, and evaluation.

The main objective was to develop a model capable of predicting future residential property prices in Tunisia. The final solution was based on two specialized XGBoost models, one for villas and one for apartments, achieving an R^2 of 0.806 for villas and 0.570 for apartments on unseen test data. The project also included uncertainty estimation through quantile regression to provide prediction intervals and better assess model reliability.

However, several limitations remain. The available data mainly represented online asking prices rather than official property valuations, and some potentially useful features such as property condition and construction year were unavailable. In addition, the forecasting component would benefit from future validation using real historical market evolution.

This internship allowed me to understand the challenges of applying machine learning to real-world data. Beyond the technical aspects, it improved my ability to work in a team, analyse problems critically, and develop solutions adapted to business needs.

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Appendices

Appendix A: Dataset Description

The 26-column shared schema used to merge all seven raw sources:

- id_annonce, source_dataset, title, description
- type_bien (property type), transaction_type (sale/rental)
- prix_tnd (price), prix_texte_brut (raw price text)
- gouvernorat (governorate), ville (city), quartier (neighbourhood)
- surface_habitable_m2 (living area), surface_terrain_m2 (land area)
- nb_pieces, nb_chambres, nb_salles_bain, nb_etages (rooms, bedrooms, bathrooms, floors)
- annee_construction, etat_bien (year built, condition — later dropped for lack of data)
- piscine, garage, jardin, climatisation, meuble, terrasse, securite (amenity flags)
- date_publication (listing publication date)

After cleaning, deduplication, and feature engineering, the final modeling dataset used 12,971 rows, split into a 10,376-row training set and a 2,595-row test set.

Appendix B: Source Code

Representative excerpt the generic source-mapping function used to reconcile all seven raw data sources into the shared schema before concatenation:

```
def mapper_dataframe(df, mapping, source_name, colonnes_finales): df = df.copy()
df.columns = df.columns.str.strip() df = df.rename(columns=mapping) df['source_dataset'] =
source_name df_final = df.reindex(columns=colonnes_finales) return df_final
```

Representative excerpt — the signature of the final deployment function (implementation detail omitted here for brevity; full logic described in Section 3.7):

```
def forecast_price(property_features: dict, years_ahead: int) -> dict: """Routes to the villa or apartment model, derives location features automatically, and returns a point estimate with a 95% interval projected years_ahead into the future.""" ...
```

Appendix C: Additional Figures and Graphs

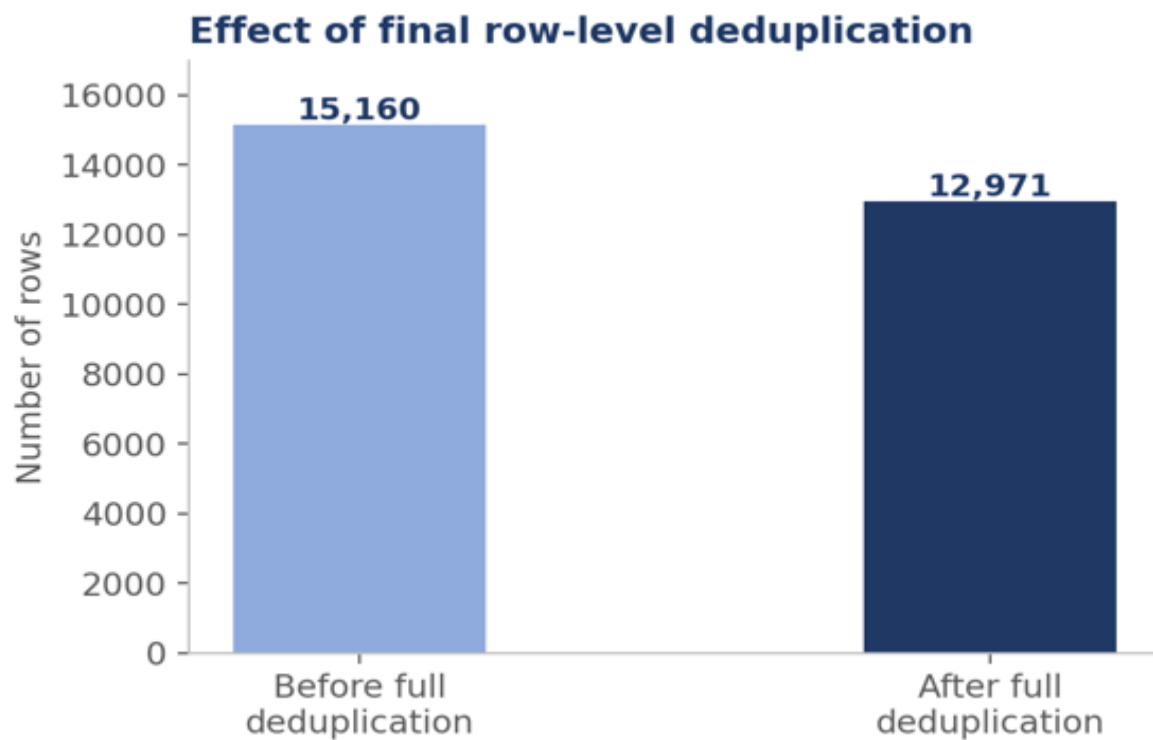


Figure C.1 — Effect of final row-level deduplication on the modeling dataset.